Laguerre and Full-Input Linearized MPC for RC-Car Simulation: A Progress Report

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https://github.com/arch1703/Laguerre_MPC_Car

***Abstract*—This report documents an autonomous RC-car simulation using Laguerre-parameterized successively linearized model predictive control (MPC), compared with a full-input linearized controller. The implementation combines local paired track-marking measurements, obstacle-aware planning, constrained optimization, nonlinear validation, and checked stopping. Independently generated obstacles are observed through a sensor interface; their future trajectories are not supplied to control. A current deterministic robustness matrix contains sixteen scenarios evaluated with both controllers, including sensor noise, latency, losses, uncertain motion classes, limited plant mismatch, and crossing objects. Both controllers satisfy mission criteria in nine of sixteen cases; all executed portions satisfy the monitored safety criteria, with no contacts or road departures. Laguerre reduces the decision count from twelve to eight, but the mean of per-run called-optimizer medians is 9.03 ms versus 4.63 ms for full-input MPC. Several uncertainty cases still stall, require fallback, or lose a verified control. An earlier, separate 23-case controlled study and selected videos provide developmental context. The evidence supports a reproducible educational comparison, not universal avoidance, recursive feasibility, hard-real-time performance, or hardware readiness.**

***Index Terms*—model predictive control, Laguerre functions, autonomous vehicle simulation, successive linearization, obstacle avoidance, reproducibility**

I. Introduction

Model predictive control repeatedly optimizes a finite sequence of inputs subject to a model and constraints. It does not require a trained driving policy. Laguerre functions provide a compact parameterization of the input sequence, potentially reducing optimization dimension [1]. However, dimension reduction alone does not establish a runtime advantage, and constrained MPC guarantees require assumptions beyond a successful simulation [2].

The project's purpose is an inspectable demonstration of Laguerre MPC for an RC-scale vehicle, with ordinary full-input linearized MPC as a matched baseline. The development progressed from oval tracking to static avoidance, independent random appearances/disappearances, moving objects, and local-marking following on multiple tracks. A full nonlinear optimizer is retained as an obstacle-free tracking baseline; it is not the controller responsible for the successful moving-obstacle demonstrations.

This report contributes an implementation and evidence snapshot rather than a new stability theorem. It describes the measurement boundary, failure semantics, matched comparisons, observed timing, retained negative results, and planned corrective work. Development is paused for publication. This document uses an IEEE-conference-style layout but is an unsubmitted, non-peer-reviewed project report; no IEEE endorsement is implied.

TABLE I
Shared Current Boundary-Experiment Settings

Quantity	Value
Sample period / target speed	0.05 s / 0.8 m/s
Wheelbase / road half-width	0.25 m / 0.85 m
Prediction / control horizon	12 / 6 samples
Steering magnitude / rate	22° / 120°/s
Acceleration range	$[-4,\ 6.67]$ m/s^2
Jerk magnitude / speed range	12 m/s^3 / $[0,\ 3]$ m/s
Laguerre modes per input / decay	4 / 0.7
QP iteration cap	100
Guarded linearization passes	Up to 6

II. Vehicle and Control Formulation

A. Discrete Kinematic Model

The rear-axle state and control are $x = [X, Y, \psi, v]^\top$ and $u = [\delta, a]^\top$. With sample period T, wheelbase L, and explicit Euler integration,

$$x_{k+1} = f(x_k, u_k) = \begin{bmatrix} X_k + T v_k \cos\psi_k \\ Y_k + T v_k \sin\psi_k \\ \psi_k + T(v_k/L)\tan\delta_k \\ v_k + T a_k \end{bmatrix}. \tag{1}$$

Plant speed is not clipped to conceal prediction error. The model omits tire slip, friction limits beyond a reference curvature-speed heuristic, and real actuator dynamics.

The body extends 0.29 m forward and 0.06 m rearward from the rear axle, with half-width 0.075 m. A conservative enclosing circle has offset $b = (0.29 - 0.06)/2 = 0.115$ m and radius $r_b = \sqrt{0.175^2 + 0.075^2} \simeq 0.1904$ m. Rectangle clearance is logged separately. Required conservative obstacle and road clearance is 0.05 m.

B. Successively Linearized MPC

The controller linearizes (1) along a nonlinear warm rollout:

$$x_{i+1} \simeq A_i x_i + B_i u_i + c_i, \qquad c_i = f(\bar{x}_i, \bar{u}_i) - A_i \bar{x}_i - B_i \bar{u}_i. \tag{2}$$

Lifted prediction matrices form a quadratic tracking objective with input-rate penalties. Position, heading and speed references follow path arc length; heading differences are wrapped consistently. Full-input MPC optimizes the two inputs at each of six control stations, giving twelve variables. Input, rate, speed, road and separating obstacle constraints are imposed.

The guarded solver limits each relinearization update to 0.12 rad steering and 1 m/s^2 acceleration. Nonlinear versus affine prediction discrepancy must remain within 0.015 m position and 0.03 rad heading. A failed optimizer or inaccurate prediction is rejected; these guards do not prove global convergence or recursive feasibility.

C. Laguerre Input-Increment Parameterization

For decay $\alpha \in (0,1)$, the first discrete Laguerre function is $\ell_1(i) = \sqrt{1-\alpha^2}\alpha^i$, $i \geq 0$. The implementation uses

$$\ell_j(0) = (-\alpha)^{j-1}\sqrt{1-\alpha^2}, \tag{3}$$

$$\ell_j(i) = \alpha\ell_j(i-1) + \ell_{j-1}(i-1) - \alpha\ell_{j-1}(i). \tag{4}$$

For each input channel, increments and absolute inputs satisfy

$$\Delta u_i = \sum_{j=1}^{n_\ell} \ell_j(i)\eta_j, \qquad u_i = u_{k-1} + \sum_{q=0}^{i} \Delta u_q. \tag{5}$$

Four modes per input give eight coefficients in the current comparison. The truncated finite-horizon basis is not assumed to be exactly orthonormal. The same costs and physical limits are applied after mapping coefficients to absolute inputs. Thus the comparison is of parameterizations, not different vehicle geometry or relaxed constraints.

III. Sensing, Planning, and Stopping

A. World–Observation Separation

A private world generates obstacle motion independently of the vehicle's control, path, feasibility and outcome. In the random-static mode, births, positions and optional lifetimes are sampled independently; the first recordings use fresh, unselected seeds. Arbitrary unsafe appearances are not postponed to obtain a successful video.

Boundary experiments use independently specified crossing motion: two transverse sinusoidal movers, one static object and one temporary static object. The base crossing period is 28 s with amplitude 1.3 m and maximum speed approximately 0.292 m/s. These are world dynamics, not a directional forecast delivered to MPC.

The sensor supplies timestamped object IDs, positions, radii, position-error bounds and admissible motion-speed bounds. It also supplies local paired boundary lines over an ideal 3.5 m range. Pose and marking association are assumed known. These are simulated measurements, not camera-derived detections. Only initialization, sensing and an independent physical monitor access world truth.

B. Local Track and Obstacle Handling

The sensed paired boundaries form an open local corridor. The planner tests center, left and right smooth detours for curvature and footprint clearance. Global track geometry is not passed to planning as a substitute for missing measurements. Oval, kidney and chicane tracks use the same controller settings.

For measured obstacle j, the controller uses a direction-free inflated radius of the form

$$R_j^{\mathrm{guard}} = R_j + \epsilon_j + m + \bar{v}_j(\tau_j + H), \tag{6}$$

where ϵ_j bounds position error, m is an optional margin, $\bar{v}_j$ is an admissible speed, τ_j is measurement age and H includes the relevant prediction or braking horizon and sample allowance. This is a reachable-region bound, not a velocity or route prediction. Unknown classes receive the movable-object bound even when physically static.

The sensor uses exactly twenty indexed random draws per acquisition tick, independent of the number of detected objects or past vehicle trajectory. Line error is a bounded per-frame coordinate translation, not independent point jitter; the corridor is shrunk by the stated bound and margin. Acquisition-frame latency preserves capture times, and the initial frame bootstraps the stream. Lost markings retain their last observed geometry.

C. Validation and Failure Semantics

Candidate controls undergo nonlinear rollout checks. A tracking action also requires a checked braking reserve from its next nominal state. The braking horizon is twenty-four samples, with terminal speed and acceleration conditions. The vehicle completes stopping before resuming tracking. Physical monitoring samples vehicle and world motion separately, including inter-sample allowances; future truth samples are not used to select commands.

With complete miss-free scans, an absent object is outside the acquired sensor footprint. It can be excluded from a local stopping tube only after checking that tube remains inside the observed region as it shrinks with measurement age and admissible object motion. This does not protect against arbitrary new objects spawning inside that region.

With Bernoulli missed detections, complete-scan exclusion is disabled. Repeated negative observations can clear memory only under the configured containment/confirmation rule, but consecutive misses can still constitute false absence evidence. No deterministic avoidance guarantee follows.

If no checked action exists, the simulation reports `no_verified_control` without advancing under an unchecked command. A stopped car is not automatically successful: stopping below 0.05 m/s for twelve seconds causes `stalled`. Current mission success requires completion, safety margins, zero fallback, and progress of at least 35% of nominal distance over the *requested* duration. Safety pass describes only the executed portion.

TABLE II
EARLIER CONTROLLED STUDY: SELECTED, SEPARATE RESULTS

Case	RMS [cm]	Solve [ms]	Step p99 [ms]
Laguerre tracking	0.130	12.78	43.84
Full-input tracking	0.142	8.29	24.58
Nonlinear tracking	0.142	836.54	925.97
Laguerre static	0.228	28.53	56.27
Full-input static	0.249	13.77	29.24
Short-horizon Laguerre static	0.227	10.76	24.08

IV. EXPERIMENTAL PROTOCOL

A. Two Distinct Evidence Sets

The earlier controlled study contains twenty-three tracking, static, appearing/replayed and parameter-sweep cases. Defaults include a 1.2 m/s target and horizons 20/10; the short-horizon static variant uses 10/5. Its results are retained separately with an implementation snapshot. They must not be pooled with the newer boundary robustness matrix.

The current matrix pairs Laguerre and full-input MPC across sixteen predeclared scenarios, requesting forty seconds per run. Settings are in Table I. Each pair shares world configuration, seed 41, initial state, costs, bounds and indexed disturbances. Observations may differ because trajectories and sensor footprints differ. Controller order alternates. Warm-up wall time is recorded separately; all in-run outliers remain included. One seed per configuration does not estimate a statistical success probability.

B. Metrics and Timing

Planned-path RMS is distance to the contemporaneous local planned route; track cross-track error is measured independently against the centerline. Obstacle detours intentionally increase centerline deviation.

Called-optimizer medians are computed only where MPC was actually invoked, excluding waiting ticks. The reported aggregate is the arithmetic mean of sixteen per-run medians, not a pooled tick median. Original all-tick solve-stage statistics are retained separately. Online time includes sensing, planning, solve and validation; full-loop time also includes physical monitoring and simulator bookkeeping. Stage percentiles are not additive.

The optional deadline watchdog selects checked braking if pre-application elapsed work already exceeds 50 ms. It cannot undo elapsed delay, and braking computation can itself overrun. Desktop measurements and iteration caps do not provide hard-real-time guarantees.

V. RESULTS

A. Earlier Controlled Study

All twenty-three cases met that study's safety/completion criteria without sampled collisions or fallback; accuracy and timing targets were not universally met. Selected values appear in Table II.

Nonlinear tracking missed all 300 deadlines in its fifteen-second experiment. It is an offline baseline, not evidence of real-time nonlinear obstacle avoidance. Reduced Laguerre dimension did not produce faster solves than full-input MPC in the measured small-QP cases.

TABLE III
CURRENT PAIRED ROBUSTNESS RESULTS

Measurement	Laguerre	Full-input
Mission pass	9/16	9/16
Executed-portion safety pass	16/16	16/16
Decision variables	8	12
Mean called-solve median [ms]	9.03	4.63
Online misses / ticks	53/11887	1/11886
Full-loop misses / ticks	213/11887	3/11886

TABLE IV
ALL SIXTEEN CONDITIONS: OUTCOME SHARED BY BOTH CONTROLLERS

Condition	Mission outcome
Nominal chicane, kidney, oval	Pass (3 conditions)
Bounded marking/object noise	Pass
Three-sample acquisition latency	Fail: fallback
Brief marking loss	Pass
Long marking loss	Fail: stalled
Missed detections	Fail: no verified control
Unknown motion classes	Fail: stalled
Tested plant mismatch	Pass
Faster crossings	Fail: rejection/stalled
Reversed crossings	Pass
Different initial station	Pass
Larger objects	Fail: fallback
Combined uncertainty	Fail: no verified control
Watchdog enabled	Pass; no intervention

B. Current Robustness Matrix

Both controllers pass nine of sixteen missions, with identical pass/fail patterns. All thirty-two executed portions satisfy monitored safety criteria, with no contacts or road departures. Table III summarizes aggregate results; Table IV retains failed conditions.

The nominal kidney pair completes approximately 1.13 laps in forty seconds. Minimum conservative obstacle clearance is 11.88 cm for both. Planned-path RMS is 2.71 cm for Laguerre and 2.70 cm for full-input MPC; full-loop p99 is 47.16 and 34.38 ms respectively. These percentiles do not imply zero outliers.

Latency causes 77 rejected QPs/fallbacks per controller; larger objects cause one each. Faster crossings cause 237 rejected QPs each and then a stall. An exit flag of -2 establishes optimizer rejection, not physical unavoidability. Compatibility of the geometric planner, finite-horizon QP and restarting state remains a working diagnosis.

The missed-detection Laguerre run ends at 35.00 s with relevant object age 17.75 s; combined uncertainty ends at 39.65 s with age 19.40 s. Growing stale reachable regions require investigation, not silent deletion. Unknown-class conservatism can legitimately block passage. Long marking loss ends in a stall before markings return. The watchdog pair records zero interventions and therefore does not exercise actual overrun recovery.

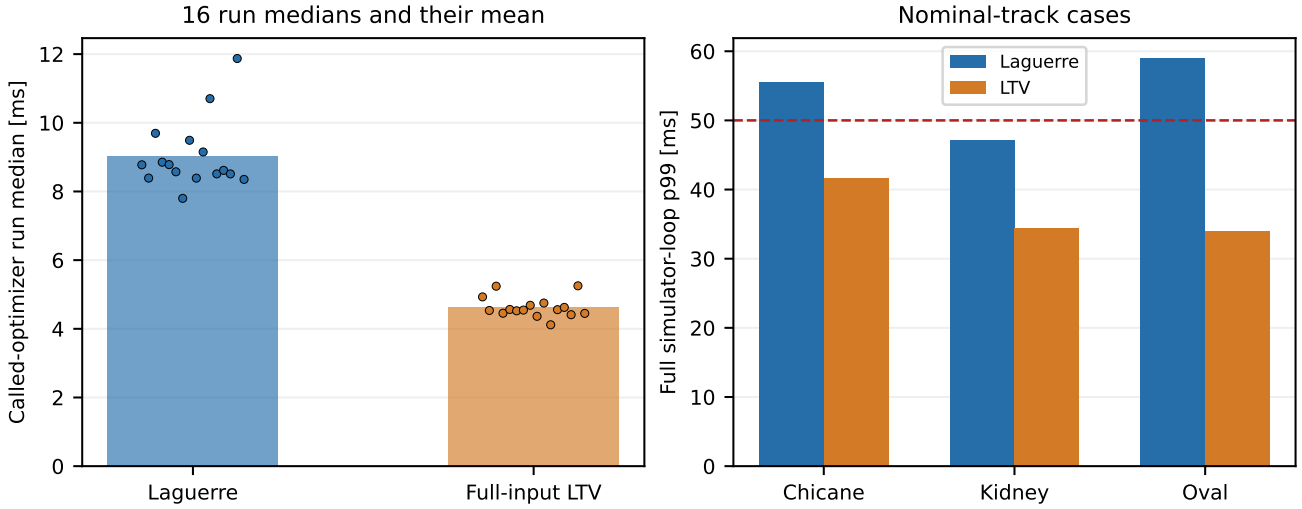


Fig. 1. Figures rebuilt from committed CSVs. Left: sixteen called-optimizer run medians per controller (points) and their arithmetic mean (bars). Right: full-loop p99 for the three nominal track cases; dashed line denotes the 50 ms sample budget. Fewer decision variables did not imply lower runtime.

C. Selected Recordings

The latest noisy-sensing chicane recording completes forty seconds and 1.025 laps, with 3.92 cm path RMS, 15.11 cm minimum conservative clearance, and no contacts or fallback. It uses the first predeclared noisy case, not a favorable seed retry. Figure 2 shows a recorded frame. Control runs at 20 Hz; the H.264 recording contains 400 frames at 10 fps.

Earlier cross-track chicane/kidney recordings demonstrate verified yielding/resumption with a previous implementation. Independent random-static seed 312882142 completes forty-five seconds without contact/fallback. The transient-object seed 273210986 is deliberately retained as a failure: seventeen births and twelve removals occur before a requested sixty-second run stops at 31.85 s on `unsafe_spawn`, without confirmed contact. That simulator halt is neither a completed mission nor demonstrated physical emergency braking. Videos and configurations from different stages are not interchangeable evidence.

VI. Reproducibility and Limitations

The public repository includes the current MATLAB/Python implementation, tests, all thirty-two current run CSV sets and manifests, case definitions, warm-up data, comparisons, checksums, paper source and selected MP4s. Private local paths are replaced by portable relative run paths, with original run IDs retained. Duplicate current source snapshots and MAT binaries are omitted. CSV animation remains supported; MATLAB's saved-bundle replay API requires an original `run.mat` or a newly saved simulation. The earlier study has a separate summary and implementation snapshot.

The controller snapshot passed 51 MATLAB regressions with zero Code Analyzer messages. Publication changes pass thirteen Python tests, including actual headless animation exports, portable paths and non-destructive curation. A prior audit checked 23,773 ticks, 81,099 world rows, 47,006 memory rows and 1,016,888 line pairs for observation causality, bounds, matched streams and stopping certificates. Timing collection preceded video rendering.

Important limitations are ideal pose/marking association, no real occlusion or camera pipeline, bounded synthetic noise, probabilistic negative evidence under missed detections, finite planning/checking horizons, limited model mismatch, one seed per condition and machine-dependent desktop timing. A car that is stationary can still be approached by an object later. Neither early termination nor measured positive clearance establishes future safety. No recursive-feasibility theorem, general robust stability proof, training-based generalization claim, ROS2 deployment or hardware test is made.

VII. Planned Work and Conclusion

The paused next-work plan prioritizes evidence-backed rejection diagnostics, dynamically reachable planner/MPC transitions, consistent time-indexed uncertainty bounds, justified memory refinement, explicit recovery versus information-limited abort, deterministic overrun/held-input-delay testing and predeclared held-out experiments. Existing clearance, progress and zero-fallback mission criteria must not be weakened to improve the pass count.

The project has achieved a reproducible simulation demonstration of Laguerre-parameterized linearized MPC for track following and obstacle handling. The matched comparison is equally important: reducing twelve inputs to eight coefficients does not deliver a measured speed advantage here. Nine of sixteen current missions pass for both variants, while retained negative results define the remaining engineering work. The appropriate current outcome is an honest simulation progress report and comparison resource, not a claim of completed general autonomous driving.

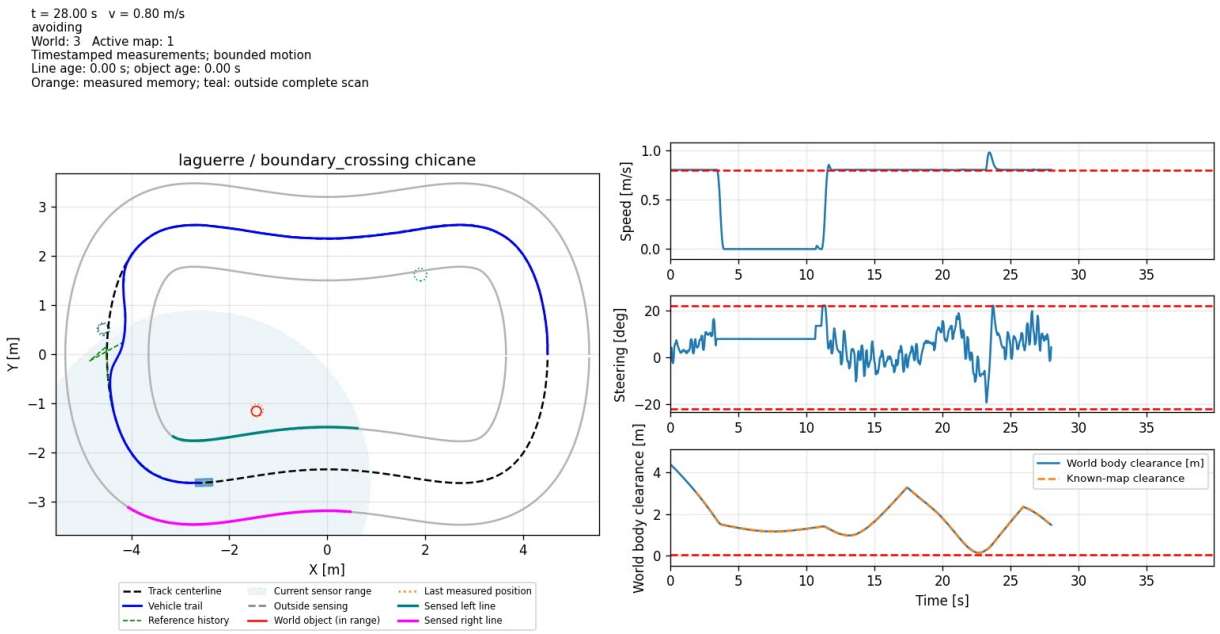


Fig. 2. Recorded noisy-sensing Laguerre simulation, not an illustrative rendering of a hypothetical outcome. Physical truth, measured memory, local markings, speed, steering and clearance are displayed separately. The README links directly to this video and the advanced cross-track recordings.

REFERENCES

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